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Case Manager Setup

In this page you can find a set of suggestions for Case Manager monitoring using Grafana.

Suggestionsâ

info

There are VM resource related metrics (CPU, Memory, IO, etc) that can also be added to Grafana. These were left out on purpose Google already has out of the box monitoring support for these resources.

These suggestions can be either new dashboards, changes to existing dashboards or completely new dashboard pages. Please find the metric description in Case Manager Top Metrics.

Cluster pageâ

Currently there is only one page where we can filter metrics by each of the instances. Although useful, having a cluster view monitoring page gives a view of how the overall cluster health. Given this, our suggestion is to add a new page with the following metrics:

Row Title	Title	Description	Query Details
Healthchecks - Case Manager	Database	The database health check relies on the shared CM's connection pool to execute the statement SELECT 1. If the connection is successfully retrieved and the statement finishes, the health check succeeds, if any error occurs in any of the steps the check fails. A timeout of 5 seconds was given when retrieving a connection to avoid hanging the health check response. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0.	**Query A** **Field:** "result" **From:** "healthchecks" **Where:** "healthcheck" = 'Database' , $timeFilter **Group By:** time($__interval), "host"
Healthchecks - Case Manager	Messaging	The Messaging Server Health Check is achieved by trying to initialize a new connection with the messaging server. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0.	**Query A** **Field:** "result" **From:** "healthchecks" **Where:** "healthcheck" = 'Messaging' , $timeFilter **Group By:** time($__interval), "host"
Healthchecks - Case Manager	Pulse	The Pulse Health Check when triggered tries to connect to pulse and tries to login. The outcome is a bool,	**Query A** **Field:** "result" **From:** "healthchecks"

Row Title	Title	Description	Query Details
		meaning that it was successfully executed when 1, otherwise is 0.	**Where:** "healthcheck" = 'Pulse' , $timeFilter **Group By:** time($__interval), "host"
Healthchecks - Case Manager	Tokenizer	The tokenizer health checks validates if there is a tokenizer instance reachable and if the authorization configured is valid. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0.	**Query A** **Field:** "result" **From:** "healthchecks" **Where:** "healthcheck" = 'Tokenizer' , $timeFilter **Group By:** time($__interval), "host"
Healthchecks - Case Manager	RabbitMQ Consumers per Event Queue	Measures the total number of consumers per event queue. Having a number of consumers less then the number of nodes represent a problem in the sense that only one/some of the casemanager(s) is/are consuming events, not distributing the load.	**Query A** **Field:** "consumers" **From:** "rabbitmq_queue" **Where:** queue =~ /*.EVENT_QUEUE.*/ , vhost =~ /hsbcasemanager/ , $timeFilter **Group By:** time($__interval), "queue", "vhost"
Case Manager End-to-End Check	End-to-End Check	Measures the execution of a operation set, constituted by 5 operations: - login - get alert list - load first alert list - get all queues - load first queue The operations are executed sequentially, and if one fails the value is immediately returned, meaning that, the set is successfully executed when the outcome is 5. It means also, that, when value is 1 for instance, the login was successfully done, but if fails getting the alert list.	**Query A** **Field:** "operations" **From:** "casemanager_monitoring" **Where:** $timeFilter **Group By:** time($__interval)
Case Manager End-to-End Check	End-to-End Check - Elapsed Time	Measures the elapsed time to execute a set of operations, per case manager instance. This graph is coupled with the "Operation Set Execution", once it shares the same operations The operations are executed sequentially, and if one fails the elapsed time will be the the time when the operation failed minus the start time.	**Query A** **Field:** "elapsed_time" **From:** "casemanager_monitoring" **Where:** $timeFilter **Group By:** time($__interval)
Event Processing - New Event	New Event - Throughput	Measures the overall throughput for the event processing when the event type is: "new event".	**Query A** **Field:** "count" **From:** "cm.event.processing.latency" **Where:** eventType='NEW_EVENT' , $timeFilter

Row Title	Title	Description	Query Details
			Group By: time($__interval), "host" **Query C** **Field:** "uptime" **From:** "system" **Where:** $timeFilter **Group By:** time($interval), "host"
Event Processing - New Event	New Event - Latency per Percentile	Measures the overall cluster event processing max latency, when the event type is "new event". It displays the higher latencies for percentile 50, 90, 99, 99.9 and also the maximum.	**Query A** **Field:** "p50", "p95", "p99", "p999" **From:** "cm.event.processing.latency" **Where:** eventType='NEW_EVENT' , $timeFilter **Group By:** time($__interval), "host" **Query B** **Field:** "max" **From:** "cm.event.processing.latency" **Where:** eventType='NEW_EVENT' , $timeFilter **Group By:** time($__interval)
Event Processing - Status Change	Status Change - Throughput	Measures the overall throughput for the event processing when the event type is: "status change".	**Query A** **Field:** "count" **From:** "cm.event.processing.latency" **Where:** eventType='STATUS_CHANGE' , $timeFilter **Group By:** time($__interval), "host"
Event Processing - Status Change	Status Change - Latency per Percentile	Measures the overall cluster event processing max latency, when the event type is "status change". It displays the higher latencies for percentile 50, 90, 99, 99.9 and also the maximum.	**Query A** **Field:** "p50", "p95", "p99", "p999" **From:** "cm.event.processing.latency" **Where:** eventType='STATUS_CHANGE' , $timeFilter **Group By:** time($__interval), "host" **Query B** **Field:** "max" **From:** "cm.event.processing.latency" **Where:** eventType='STATUS_CHANGE' , $timeFilter **Group By:** time($__interval)
Automation Rules	Automation Rules - Throughput and Errors	Measures the overall throughput and errors when executing automation rules.	**Query A** **Field:** "count" **From:** "cm.automation" **Where:** $timeFilter **Group By:** time($__interval), "host", "ruleName"
Automation Rules	Automation Rules - Latency per Percentile	Measures the overall cluster latency at percentile 99 to execute an automation rule. It displays highest automation rule latency at percentile 99.9 per instance.	**Query A** **Field:** "p999" **From:** "cm.automation" **Where:** $timeFilter **Group By:** time($__interval), "host", "ruleName"
Pulse Requests	Pulse Requests - Throughput and Errors	Measures the overall throughput and the errors in the communication with pulse instances.	**Query A** **Field:** "count" **From:** "cm.request.pulse.api" **Where:** $timeFilter **Group By:** time($__interval), "host", "path"
Pulse Requests	Pulse HTTP Endpoints and Latencies	Measures the latency in the communication with pulse instances.	**Query A** **Field:** "count", "p99" **From:** "cm.request.pulse.api" **Where:** $timeFilter **Group By:** "host", "path"
Event Processing	Event Processing Queue Occupancy	Measures the event processing queue occupancy	**Query A** **Field:** "value"

Row Title	Title	Description	Query Details
		(internal cm queue), per case manager instance.	**From:** "cm.event.processing.queue.ratio" **Where:** $timeFilter **Group By:** time($__interval), "host"
Event Processing	Event Processing Batch audit-batch Queue Occupancy	Measures the audit batch queue occupancy (internal cm queue), per case manager instance.	**Query A** **Field:** "value" **From:** "cm.event.processing.queue.occupancy.audit-batch" **Where:** $timeFilter **Group By:** time($__interval), "host"
Event Processing	Event Processing Batch audit-batch Flush Latency at p99	Measures the audit batch queue latency (internal cm queue), per case manager instance.	**Query A** **Field:** "p99" **From:** "cm.event.processing.batch.flush.latency.audit-batch" **Where:** $timeFilter **Group By:** time($__interval), "host"
Event Processing	Event Processing Batch explanations-batch Queue Occupancy	Measures the explanations batch queue occupancy (internal cm queue), per case manager instance.	**Query A** **Field:** "value" **From:** "cm.event.processing.queue.occupancy.explanations-batch" **Where:** $timeFilter **Group By:** time($__interval), "host"
Event Processing	Event Processing Batch explanations-batch Flush Latency at p99	Measures the explanations batch queue latency (internal cm queue), per case manager instance.	**Query A** **Field:** "p99" **From:** "cm.event.processing.batch.flush.latency.explanations-batch" **Where:** $timeFilter **Group By:** time($__interval), "host"
Event Processing	Event Processing Batch event-storage-batch Queue Occupancy	Measures the event storage batch queue occupancy (internal cm queue), per case manager instance.	**Query A** **Field:** "value" **From:** "cm.event.processing.queue.occupancy.event-storage-batch" **Where:** $timeFilter **Group By:** time($__interval), "host"
Event Processing	Event Processing Batch event-storage-batch Flush Latency at p99	Measures the event storage batch queue latency (internal cm queue), per case manager instance.	**Query A** **Field:** "p99" **From:** "cm.event.processing.batch.flush.latency.event-storage-batch" **Where:** $timeFilter **Group By:** time($__interval), "host"
Event Processing	Event Processing Batch event-common-batch Queue Occupancy	Measures the event common batch queue occupancy (internal cm queue), per case manager instance.	**Query A** **Field:** "value" **From:** "cm.event.processing.queue.occupancy.event-common-batch" **Where:** $timeFilter **Group By:** time($__interval), "host"
Event Processing	Event Processing Batch event-common-batch Flush Latency at p99	Measures the event common batch queue latency (internal cm queue), per case manager instance.	**Query A** **Field:** "p99" **From:** "cm.event.processing.batch.flush.latency.event-common-batch" **Where:** $timeFilter **Group By:** time($__interval), "host"
Event Processing	Event Processing Batch event-queue-automations-batch Queue Occupancy	Measures the event queue automations batch queue occupancy (internal cm	**Query A** **Field:** "value" **From:** "cm.event.processing.queue.occupancy.event-queue-automations-batch"

Row Title	Title	Description	Query Details
		queue), per case manager instance.	**Where:** $timeFilter **Group By:** time($__interval), "host"
Event Processing	Event Processing Batch event-queue-automations-batch Flush Latency at p99	Measures the event queue automations batch queue latency (internal cm queue), per case manager instance.	**Query A** **Field:** "p99" **From:** "cm.event.processing.batch.flush.latency.event-queue-automations-batch" **Where:** $timeFilter **Group By:** time($__interval), "host"
Deadletters	Deadletter Events	Measures the total number of deadletters.	**Query A** **Field:** "count" **From:** "cm.event.processing.deadletter.count" **Where:** $timeFilter **Group By:** time($__interval), "host"
DB Connection Pool	DB Connection Pool usage	Measures the overall db connection pool usage.	**Query A** **Field:** "value" **From:** "cm.db.connection.pool.usage" **Where:** $timeFilter **Group By:** time($__interval), "host"
DB Connection Pool	DB Connection Pool Waiting Threads	Measures the overall db connection pool waiting threads.	**Query A** **Field:** "count" **From:** "cm.db.connection.pool.waiting" **Where:** $timeFilter **Group By:** time($__interval), "host"
Case Manager HTTP Endpoints and Latencies	Case Manager HTTP Endpoints and Latencies	Measures the latency of Case Manager HTTP endpoints.	**Query A** **Field:** "count", "p99" **From:** "cm.request.api" **Where:** $timeFilter **Group By:** "host", "path"
Login Errors	Login Error Percent(10 Minute Window)	Measures the number of logins and its error percentage over a 10 minute window.	**Query A** **Field:** "value" **From:** "cm.request.login" **Where:** $timeFilter **Group By:** time($__interval), "host"
JVM	JVM Heap Usage	Measures the jvm heap usage per component instance.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.memory.heap.usage" **Where:** $timeFilter **Group By:** time($__interval), "host"
JVM	JVM Heap Usage After GC	Measures the jvm heap usage after gc per instance.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.memory.heap.after-gc.usage" **Where:** $timeFilter **Group By:** time($__interval), "host"
JVM	JVM Garbage Collector Time	Measures maximum JVM GC duration. Long garbage collection pause durations indicate that the heap size is not adjusted for the actual usage.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.gc.G1-Old-Generation.time" **Where:** $timeFilter **Group By:** "host"
JVM	JVM Active Threads Count	Measures the current number of active threads in the JVM per instance.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.threads.count" **Where:** $timeFilter **Group By:** time($__interval), "host"
JVM			

Row Title	Title	Description	Query Details
	JVM File Descriptors Ratio	Measures the jvm file descriptors usage ratio per instance.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.filedescriptor" **Where:** $timeFilter **Group By:** time($__interval), "host"

The `Row Title` column refers to a Grafana feature to group dashboards by type. This is only a suggestion and thus not mandatory.

Node pageâ

These suggestions relate to the existing Case Manager dashboard page. You may find some new dashboards or changes to existing ones.

Our suggestions are to add the dashboards in the following table with a filter by each of the available nodes and applications:

Row Title	Title	Description	Query Details
Case Manager - Healthchecks	Database	The database health check relies on the shared CM's connection pool to execute the statement SELECT 1. If the connection is successfully retrieved and the statement finishes, the health check succeeds, if any error occurs in any of the steps the check fails. A timeout of 5 seconds was given when retrieving a connection to avoid hanging the health check response. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0. This healthcheck corresponds to the case manager selected instance.	**Query A** **Field:** "result" **From:** "healthchecks" **Where:** "healthcheck" = 'Database' , "host" = '$host' , $timeFilter **Group By:** time($__interval)
Case Manager - Healthchecks	Messaging	The Messaging Server Health Check is achieved by trying to initialize a new connection with the messaging server. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0. This healthcheck corresponds to the case manager selected instance.	**Query A** **Field:** "result" **From:** "healthchecks" **Where:** "healthcheck" = 'Messaging' , "host" = '$host' , $timeFilter **Group By:** time($__interval)
Case Manager - Healthchecks	Pulse	The Pulse Health Check when triggered tries to connect to pulse and tries to login, if it succeeds the check will also succeed. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0. This healthcheck corresponds to the case manager selected instance.	**Query A** **Field:** "result" **From:** "healthchecks" **Where:** "healthcheck" = 'Pulse' , "host" = '$host' , $timeFilter **Group By:** time($__interval)
	New Event - Throughput	Measures the event processing throughput when the event type	**Query A** **Field:** "count"

Row Title	Title	Description	Query Details
Event Processing - New Event		is "new event", for the selected case manager instance.	**From:** "cm.event.processing.latency" **Where:** "host" = '$host' , eventType='NEW_EVENT' , $timeFilter **Group By:** time($__interval)
Event Processing - New Event	New Event - Latency per Percentile	Measures the event processing latencies when the event type is "new event", for the selected case manager instance.	**Query A** **Field:** "p50", "p95", "p99", "p999", "max" **From:** "cm.event.processing.latency" **Where:** "host" = '$host' , eventType='NEW_EVENT' , $timeFilter **Group By:** time($__interval)
Event Processing - Status Change	Status Change - Throughput	Measures the event processing throughput when the event type is "status change", for the selected case manager instance.	**Query A** **Field:** "count" **From:** "cm.event.processing.latency" **Where:** "host" = '$host' , eventType='STATUS_CHANGE' , $timeFilter **Group By:** time($__interval)
Event Processing - Status Change	Status Change - Latency per Percentile	Measures the event processing latencies when the event type is "status change", for the selected case manager instance.	**Query A** **Field:** "p50", "p95", "p99", "p999", "max" **From:** "cm.event.processing.latency" **Where:** "host" = '$host' , eventType='STATUS_CHANGE' , $timeFilter **Group By:** time($__interval)
Automation Rules	Automation Rule $automationRule - Throughput and Error Percent	For the selected case manager instance, measures the throughput to execute the selected automation rule and the associated error percentage.	**Query A** **Field:** "count" **From:** "cm.automation" **Where:** "host" = '$host' , "ruleName" = '$automationRule' , $timeFilter **Group By:** time($__interval)
Automation Rules	Automation Rule $automationRule - Latency per Percentile	For the selected case manager instance, measures the latency per percentile (50, 95, 99, 99.9 and 100) to execute the selected automation rule.	**Query A** **Field:** "p50", "p95", "p99", "p999", "max" **From:** "cm.automation" **Where:** "host" = '$host' ruleName = '$automationRule' , $timeFilter **Group By:** time($__interval)
Pulse Requests	Pulse Requests - Throughput and Error Percent	For the selected case manager instance, measures the throughput of the communications with pulse and the error percent inherent to this.	**Query A** **Field:** "count" **From:** "cm.request.pulse.api" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval), "path"
Pulse Requests	Pulse HTTP Endpoints and Latencies	Measures the latency in the communication with pulse instances.	**Query A** **Field:** "count", "p99" **From:** "cm.request.pulse.api" **Where:** "host" = '$host' , $timeFilter **Group By:** "path"
Event Processing	Event Processing Queue Occupancy	Measures the percentage of the event processing queue being used, for the selected case manager instance.	**Query A** **Field:** "value" **From:** "cm.event.processing.queue.ratio" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval)
Event Processing	Events Waiting in Queue	Measures the number of events waiting to be processed, for the selected case manager instance.	**Query A** **Field:** "value" **From:** "cm.event.processing.queue.used.count" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval)

Row Title	Title	Description	Query Details
Deadletters	Deadletter Events	Measures the number of events that were selected case manager deadletter.	**Query A** **Field:** "count" **From:** "cm.event.processing.deadletter.count" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval)
DB Connection Pool	DB Connection Pool Usage	Measures the db connection pool usage, for the selected case manager instance.	**Query A** **Field:** "value" **From:** "cm.db.connection.pool.usage" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval)
DB Connection Pool	DB Connection Pool Waiting Threads	Measures the db connection pool waiting threads, for the selected case manager instance.	**Query A** **Field:** "count" **From:** "cm.db.connection.pool.waiting" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval)
Case Manager Executor Thread Pool and Task Queue	Executor Thread Pool Usage	Measures the thread pool usage, for the selected case manager instance.	**Query A** **Field:** "value" **From:** "cm.executor.pool.threadsUsageRatio" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval)
Case Manager Executor Thread Pool and Task Queue	Tasks on Executor's Queue	Measures the the number of tasks within the executor's queue, on the selected case manager instance.	**Query A** **Field:** "value" **From:** "cm.executor.pool.taskQueueEventsCount" **Where:** "host" = '$host' $timeFilter **Group By:** time($__interval)
Case Manager HTTP Endpoints and Latencies	Case Manager HTTP Endpoints and Latencies	Measures the latency of Case Manager HTTP endpoints.	**Query A** **Field:** "count", "p99" **From:** "cm.request.api" **Where:** "host" = '$host' , $timeFilter **Group By:** "path"
Login Errors	Login Error Percent	Measures the login error percent on the selected case manager instance.	**Query A** **Field:** "value" **From:** "cm.request.login.error.percent" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval)
JVM	JVM Heap	Measures the jvm heap usage of the selected instance.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.memory.heap.used" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval) **Query B** **Field:** "value" **From:** "letmeknow.jvm.memory.heap.max" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval), "host"
JVM	JVM Heap Used (By Gen)	For the selected instance, considering the heap used, displays the usage of each generation.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.memory.pools.G1-Old-Gen.used" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval)
JVM	JVM Heap Used After GC (By Gen)	For the selected instance, considering the heap used, displays the usage of each generation after gc.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.memory.pools.G1-Old-Gen.used-after-gc"

Row Title	Title	Description	Query Details
			Where: "host" = '$host' , $timeFilter **Group By:** time($__interval)
JVM	JVM Heap Usage After GC	Measures the jvm heap usage after gc on the selected instance.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.memory.heap.after-gc.usage" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval)
JVM	JVM Garbage Collector Time	Measures the JVM GC duration on the selected instance. Long garbage collection pause durations indicate that the heap size is not adjusted for the actual usage.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.gc.G1-Old-Generation.time" **Where:** "host" = '$host' , $timeFilter **Group By:** N/A
JVM	JVM Active Threads Count	Measures the current number of active threads in the JVM on the selected instance.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.threads.count" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval)
JVM	JVM File Descriptors Ratio	Measures the jvm file descriptors usage ratio of the selected instance.	**Query A** **Field:** "value" **From:** "letmeknow.jvm.filedescriptor" **Where:** "host" = '$host' , $timeFilter **Group By:** time($__interval)